ITDS 124 Mathematics of Computing

Chapter 1. Fundamentals

1.4 (Part 2) Base Numbers

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Representations of Integers

- If $b > 1$ is an integer, then every positive integer n can be uniquely expressed in the form

$$n = d_k b^k + d_{k-1} b^{k-1} + \cdots + d_1 b + d_0$$

  Where $0 \leq d_i < b,\ i = 0,1, \ldots, k$, and $d_k \neq 0$. The sequence $d_k d_{k-1} \ldots d_1 d_0$ is called the base b expansion of n.

- If we need to explicitly indicate the base b, we will write the above sequence as $(d_k d_{k-1} \ldots d_1\ d_0)_b$

Representations of Integers

- Examples
  - $5999 = 5 * 10^3 + 5 * 10^2 + 5 * 10^1 + 5 * 10^0$ (Normal number is base 10)
  - $111111_2 = 1 * 2^5 + 1 * 2^4 + 1 * 2^3 + 1 * 2^2 + 1 * 2^1 + 1 * 2^0 = 63$
  - $2132_4 = 2 * 4^3 + 1 * 4^2 + 3 * 4^1 + 2 * 4^0 = 158$
  - $2132_7 = 2 * 7^3 + 1 * 7^2 + 3 * 7^1 + 2 * 7^0 = 758$

Various bases

Base	Digits Used
2	0, 1
3	0, 1, 2
4	0, 1, 2, 3
5	0, 1, 2, 3, 4
…	0, 1, 2, 3, 4, …

Base conversion

- Find the base 4 representation of 158

$$
\begin{array}{r|l l}
4 & 158 & 2 \\
4 & 39 & 3 \\
4 & 9 & 1 \\
4 & 2 & 2 \\
 & 0 &
\end{array}
$$

- $158 = (2132)_4$

Base 2 number

- Base 2 (binary) number 11100 (or 11100_2)

is equivalent to ___?____ in base 10.

Base 3 number

- Base 3 number 12100 (or 12100_3)

is equivalent to ___?____ in base 10.

is equivalent to ___?____ in base 2.

Base number conversions

- Convert each of the following numbers to base 10:

412_6 =

374_9 =

1432_5 =

Base number conversions

- Convert each of the following base 10 numbers to the base stated:

472_{10} to base 4 =

179_{10} to base 7 =

342_{10} to base 3 =

Adding Binary Numbers

- "Carrying" concept → same as base 10

Base 10: 12,345 + 67,899 =

Base 2:

111 + 100 =

101 + 110 =

1111 + 111 =

Subtracting Binary Numbers

- "Borrowing" concept → same as base 10

Base 10: 112,345 - 67,899 =

Base 2:

111 - 101 =

110 - 11 =

1100 - 101 =

Multiplying Binary Numbers

- Multiplying by numbers like 10, 100 and 1000 is very similar to working with base 10 numbers.

Calculate the binary numbers:

1011 x 100 =

110110 x 1000 =

11011 x 10000 =

Multiplying Binary Numbers

- Calculate the binary numbers: 1011 x 11 =

```
        1 0 1 1
X         1 1
```

Multiplying Binary Numbers

- Calculate the binary numbers:

1110 x 101 =

11011 x 111 =

11011 x 1001 =

What about Division?

- Dividing binary numbers:

100001 / 11 = (? / ? In base 10) =

100001 / 100 = (? / ? In base 10) =

Long-division form

```
      ______1011___
11 )  100001
        11
         100
          11
           11
           11
            0
```

```
        _____1000___
100 )  100001
        100
          0001
```

Dividing binary numbers

- Try 111000 / 1011

Addition

- Base 5: 14 + 21 =
- Base 7 16 + 32 =
- Base 5 141 + 104 =
- Base 3 212 + 121 =

Multiplication

- Base 5: 141 x 23 =
- Base 3: 122 x 12 =
- Base 6: 512 x 24 =

Division

$2252_7 \div 23_7 = ?_{10} \div ?_{10} =$

$\mathbf{1 \times 23_7 = 23_7}$
$\mathbf{2 \times 23_7 = 46_7}$
$\mathbf{3 \times 23_7 = 102_7}$
$\mathbf{4 \times 23_7 = 125_7}$
$\mathbf{5 \times 23_7 = 151_7}$
$\mathbf{6 \times 23_7 = 204_7}$

$$
\begin{array}{r}
__66__ \\
23_7 \;\big)\; 2252_7 \\
\underline{204} \\
212 \\
\underline{204} \\
5
\end{array}
$$

Division

- Let's try $233_5 \div 13_5 = ?_{10} \div ?_{10} =$

$\mathbf{1 \times 13_5} = ?_5 =$
$\mathbf{2 \times 13_5} = ?_5 =$
$\mathbf{3 \times 13_5} = ?_5 =$
$\mathbf{4 \times 13_5} = ?_5 =$

$$\overline{}$$

$$13_5 \;)\; 233_5$$

Applications

- The bases 2, 8 and 16 are frequently used in computer science, and the base 26 is sometimes used in **cryptology** (the science of producing and deciphering secret codes).

- To use 26 as a base, we can use the letters $A, B, \ldots, Z$ to represent the digits $0,1, \ldots, 25$. In this way, we can interpret any text as the base 26 representation of an integer.

- Ex. "TWO" in base 26 $= (\mathrm{T} * 26^2) + (\mathrm{W} * 26^1) + \mathrm{O}$

$$= (19 * 676) + (22 * 26) + 14$$

$$= 13430$$

Bacon's code (1)

- As an example of cryptology, we consider a remarkable code due to Sir Francis Bacon.
- Write base 2 representation of the position of each letter of the message in the English alphabet
  - A = 0 in base 10 → 00000 in base 2
  - B = 1 in base 10 → 00001 in base 2
  - C = 2 in base 10 → 00010 in base 2
  - …
  - Z = 25 in base 10 → 11001 in base 2

Table to show the binary representation from A to Z

Alphabet	Decimal	Binary	Alphabet	Decimal	Binary
A	0	00000	N	13	01101
B	1	00001	O	14	01110
C	2	00010	P	15	01111
D	3	00011	Q	16	10000
E	4	00100	R	17	10001
F	5	00101	S	18	10010
G	6	00110	T	19	10011
H	7	00111	U	20	10100
I	8	01000	V	21	10101
J	9	01001	W	22	10110
K	10	01010	X	23	10111
L	11	01011	Y	24	11000
M	12	01100	Z	25	11001

Bacon's code (2)

- Suppose we wish to encode a message "FLEE NOW"
- Each letter will be represented by its binary form.

F	L	E	E	N	O	W
00101	01011	00100	00100	01101	01110	10110

- Next find a "dummy" message with exactly 5 times as long (you can pad with a few extra letters if necessary). Let's assume that our dummy message is

  ONCE UPON A TIME IN THE WEST THERE WAS A TOWN

Bacon's code (3)

- Then place that message in the third row as follows

F	L	E	E	N	O	W
00101	01011	00100	00100	01101	01110	10110
ONCEU	PONAT	IMEIN	THEWE	STTHE	REWAS	ATOWN

- Then each letter that matches with 1, adjust it to be italic to make the final encoded message.

F	L	E	E	N	O	W
00101	01011	00100	00100	01101	01110	10110
ONC*EU*	P*ONAT*	IM*E*IN	TH*E*WE	*STTHE*	R*EWAS*	*ATOW*N

- Final message : ONCE *UPON A TIME* IN TH*E* WES*T THERE WAS A TOW*N

Bacon's code (4)

- Overall steps of Bacon's encoding process.

Original message:

F	L	E	E	N	O	W
00101	01011	00100	00100	01101	01110	10110
ONCEU	PONAT	IMEIN	THEWE	STTHE	REWAS	ATOWN
ONCE*U*	P*ON*A*T*	IM*E*IN	TH*E*WE	S*TT*H*E*	R*EWA*S	*ATOW*N

Encoded message:

ONCE *UPON A TIME* IN TH*E* WES*T THERE WAS A TOW*N

Bacon's decoding process (1)

- Suppose that we wish to decode the following encoded message using the Bacon code process.

  *N*OW *IS* *T*HE *TI*ME FOR *A*LL G*OO*D M*EN* T*O* AI*D* T*HE* *C*OUNTR*Y*

- Since there are 40 letters, the true (original) message must have 8 letters. So we rearrange the message into a group of 5 each. Then looking at each character to map back to 0 (normal) or 1 (italic).

  *N*OW*IS* *T*HE*TI* MEFOR *A*LLG*O* *O*DM*EN* T*O*AI*D* T*HEC*O UNTR*Y*

  10010 10011 00000 10001 10011 01001 01110 00001

Bacon’s decoding process (2)

- Then convert the group of binary numbers back to characters.
- They are 18, 19, 0, 17, 19, 9, 14, 1.
- Decoded message : START JOB

Bacon's decoding process (3)

- Overall steps of Bacon's decoding process.

Encoded message :

*N*OW *I*S *T*HE *TI*ME FOR *A*LL G*OO*D M*EN* T*O* AI*D* T*HE C*OUNTR*Y*

40 letters → the true message must have 40/5 = 8 letters

*N*OW*IS* *T*HE*TI* MEFOR *A*LLG*O* *O*DM*EN* T*O*AI*D* T*HEC*O UNTR*Y*

10010	10011	00000	10001	10011	01001	01110	00001
18	19	0	17	19	9	14	1
S	T	A	R	T	J	O	B

Decoded message (or original message) :

START JOB